QP CODE: **MG3DSESDV200** Reg No: _____

Name: _____

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

THIRD SEMESTER MGU-UGP (HONOURS) REGULAR

EXAMINATION OCTOBER 2025

Third Semester

Discipline Specific Major Course - MG3DSESDV200 - ACOUSTICS AND PSYCHOACOUSTICS

(2024 Admission Onwards)

Duration: 1.5 Hours Maximum Marks: 70

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain] [CO No(s)]

Part A – Very Short Answer Questions

Answer any eight of the following. Each question carries 2 marks.

1. Define Haas effect. [U] [CO1]
2. What are combination tones? [U] [CO1]
3. State inverse square law. [K] [CO2]
4. What is a room mode? [U] [CO2]
5. Write any two materials with high absorption coefficient. [R] [CO4]
6. Define reverberation time. [K] [CO4]
7. What is NC (Noise Criterion)? [K] [CO3]
8. Define ILD and its role in localization. [U] [CO1]
9. What is the purpose of using Helmholtz resonator? [A] [CO3]
10. What is the significance of SAC or NRC? [U] [CO3]

$(8 \times 2 = 16)$

Part B – Short Essay Questions

Answer any five of the following in one paragraph each. Each question carries 6 marks.

11. Describe the perception of direction using ITD and ILD. [U] [CO1]
12. Explain standing waves and their impact on room acoustics. [An] [CO2]
13. Describe flutter echo with appropriate sketch. [U] [CO2]
14. Explain noise transmission and sound isolation techniques in rooms. [An] [CO3]
15. What are porous and diaphragmatic absorbers? [U] [CO3]

16. How is reverberation time measured in performance spaces? [An] [CO4]

17. Write short notes on absorption coefficients of commonly used materials. [U] [CO4]

$(5 \times 6 = 30)$

Part C – Essay Questions

Answer any two of the following. Each question carries 12 marks.

18. With diagrams, explain how room shape affects sound reflection and diffusion. Include convex, concave, and QRD panels. [An] [CO3]

19. Discuss in detail the psychoacoustic principles of perception of frequency, loudness, and space. [An] [CO1]

20. Explain reverberation design for different room types. Add acoustical features and case studies of at least two auditoriums. [E] [CO4]

$(2 \times 12 = 24)$

END OF QUESTION PAPER

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